

Stars and Planets in Hobart 2026: preliminary program

	Monday	Tuesday	Wednesday	Thursday	Friday	
9:00	Chair: Amanda Karakas Kateryna Andrych (review) <i>Across wavelengths and spatial scales: multiwavelength studies of second-generation protoplanetary discs</i>	Chair: Amanda Karakas Kateryna Andrych (review) <i>Across wavelengths and spatial scales: multiwavelength studies of second-generation protoplanetary discs</i>	Chair: Mark Wardle Rajika Kuruwita (review) <i>Attempting the find the origin of FUor outbursts starting from ISM scales</i>	Chair: Simon Murphy May Gade Pedersen (review) <i>Multicolour photometry of pulsating early-type stars from TESS, Gaia, and Roman</i>	Chair: Babatunde Akinsanmi Jean-Philippe beaulieu (review) <i>NASA Roman : mass function of cold planets down to the mass of Ganymede, and Free-Floating planets</i>	
9:15						
9:30		Maksym Mohorian ALMA's sharpest view of IRAS 08544-4431: Unveiling the dust distribution asymmetries in the post-AGB circumbinary disc	Bradley Johnson The interstellar lifecycle: Using dense gas to study pre-stellar clusters in the Milky Way	Joel Ong Weighing the Cores of Red Giants and Black-hole Stars using Momentum Conservation	Roslyn Forecast Teaching Exoplanet Atmospheres	
9:45		Peter Scicluna Pre-supernova dust production in the mid-infrared	Arunima Arunima A kinematic age for PDS 70 and the resulting constraints on planet formation	Lorenzo Martinelli Prospects for constraining angular momentum transport in solar-like main-sequence stars with PLATO	Tom Evans-Soma Can we determine if most sub-Neptunes are water worlds or gas dwarfs?	
10:00		Toon De Prins Gravitas axem torque: Time-series VLTi constraints on the dynamic dusty disc rim around a highly evolved binary	Jonah Goldfine Machine-learned Protoplanetary Disk Representations for JWST Interferometric Imaging	Dorsa Majidi Fourier Analysis of Gravity-Mode Period Spacing Patterns as an Age Diagnostic	Kelsea Haste TESS and JWST Phase-curve of an ultra-hot Jupiter	
10:15		Ali Pourmand Second-Generation Planets or Ablated Stars? Formation Scenarios for Pulsar Planets	Catherine Smith Long Term Variability of Dipper Stars	Tom Love Why do some stars achieve runaway pulsation amplitudes?	Owen Downes Probing The Longitudinal Structure of KELT-25b Through Atmospheric Ionisation	
10:30						
10:45	10:50 Welcome	Morning tea	Morning tea	Morning tea	Morning tea	
11:00						
11:15	Chair: Andrew Cole Sven Buder (review) <i>Precision stellar spectroscopy: extracting more from every spectrum</i>	Chair: Maksym Mohorian Taïssa Danilovich (review) <i>Lifting the veil on evolved stars with observational astrochemistry</i>	Chair: Maja Jablonska Alexander Wallace (review) <i>Combining RV and astrometry: prospects for Gaia DR4</i>	Chair: Erica Thygesen Adam Rains (review) <i>K/M Dwarfs and the Pathway to Unlocking Their Chemistry for Galactic Archaeology and Exoplanet Demographics</i>	Chair: Angharad Weeks Rebecca Nealon (review) <i>Finding misaligned planets</i>	
11:30	Giada Casali [C/N] as a chemical clock: strengths and limitations	Riley Thai Mass transfer stability on the TP-AGB	Thavisha Dharmawardena Beyond Parallax: Variable-Star Distances with Gaia DR4	Chelsea Huang Planet demographics with TESS	Ben Montet Solar System Science with TESS	
11:45	Ethan Bull Improved Chemical Abundances for AAOmega Spectra through Machine Learning	Kayla Martin The Light Common Envelope ("CE-Lite"): Tackling the Post-Giant Binary Puzzle	Simon Ellingsen Stars, Planets and Parallaxes	Zachariah Jones Exoplanet Detection Pipeline for Signals with Large Transit Timing Variations	Alexander Heger How asteroids collide	
12:00	Anell Cornejo-Cárdenas Bridging observations and simulations through chemical abundance patterns in dwarf galaxies	Harshit Pal Stellar Fingerprints: What isotopic ratios reveal about post-AGB stars	Erica Thygesen The Nyx Catalogue: Paving the way for Roman with extinction mapping of the Galactic Bulge	Alice Ellerton The Eccentricity Distribution of TESS Multi-Planet Systems	Babatunde Akinsanmi A view of the Ultra-hot Jupiter WASP-103b from its JWST/NIRSpec- Prism phasecurve	
12:15	Simon Correrá Distinguishing Different Galactic Accretion Events using GALAH DR4 Chemical Abundances	Claudia Fava Tracing the Chemistry of Evolved Stars from the Photosphere to the Circumstellar Environment	Angharad Weeks First Detection of Stellar Granulation and Oscillations from JWST Exoplanet Light Curves	Thomas Plunkett The TASSIE Program II: Three Close-In Companions Orbiting Sun-Like Stars	Ylenia Mascolo Atmospheric and evolutionary characterisation of K2-18 b	
12:30	Lunch	Lunch	Lunch and steering committee meeting	Lunch	Lunch	
12:45						
13:00						
13:15						
13:30						
13:45	Chair: Adam Rains Maja Jablonska An Empirical Map of the Galactic Bulge: Orbits, Abundances and Ages, Star by Star	Chair: Rajika Kuruwita Jade Powell Core-collapse supernova simulations with different masses, rotation rates and equations of state.	Chair: Chelsea Huang Benjamin Pope The Lazuli Space Observatory: Stellar and Exoplanet Science	Chair: Rebecca Nealon Andrew Cole OGLE-2015-BLG-1395Lb: Cold Sub-Saturn Around and M Dwarf Discovered from Tasmania	Chair: Ben Montet Alex Parsons-Nield Simulating Stargrazing Exocomets in the Beta Pictoris System	
14:00	Daniel Zucker From Little Things Big Things Grow: Deciphering the Milky Way's Accretion History with S5 and OzArch	Gang Li PLATO-TIDES: Probing binary tidal interactions in the PLATO era	Tiasha Biswas Exoplanets in Radio - using Statistics	Belinda Nicholson Project SWIFT: Exoplanet science through partnerships with space industry	Karen Pollard Studies of young star star systems which host exocomets	
14:15	Clare Worley Investigating the chemical enrichment history of the solar neighbourhood	Ryosuke Hirai Changing the mindset for rapid stellar evolution modelling	Evgeni Grishin Hot Jupiter formation by Galactic perturbations to old wide binary stars	Qichen Huang Polarised radio emission from two newly found T9 dwarfs	Xuan Che Neo Orbital Characterisation of Hyperbolic Bolides for Interstellar Candidate Identification	
14:30	Adithya Gudalur Balasubramaniam The S5 Survey's View of NGC 1851's Stellar Stream	Matthew Pinson Tracing the history of blue stragglers	Alex Andersson A volume-limited radio survey of southern active M dwarfs	Hayley Wagner Understanding Inner Super Earths and Cold Jupiters Relationship With TESS and AAPS	Michael Elston Automatic Detection of Exocomets in the β Pictoris system	
14:45	Ankita Mondal Tracing cosmic helium abundances using K-dwarfs	Muskaan Monteiro Modelling Carbon Deficient Giants: Stellar Binaries Produce Peculiar Evolved Stars	Mia Babatsikos Chemical signatures of planet engulfment in solar twins	Hayden Greer A Data-Driven Model of HST/NICMOS for High-Resolution and High-Contrast Imaging	Daniel Price Conference summary: stars and planets in Australia and where we go from here	
15:00	Diego Ulloa-Solis Self-consistent temperatures and luminosities for giant stars in globular clusters.	Ethan Hallam The Dance of Spinning Stars	Gabriel McMahon-Killingsworth Can life exist in triple star systems?	Milo Langker Spectral Information Content of Diffraction-Limited Images		
15:15	LJ Spriggs Sketching an Evolving Map of the Milky Way	Chunliang Mu Resolving the Photosphere in 3D Common Envelope Simulations with Adaptive Particle Refinement	Siska-Lut Buelens Timing General Relativity: Searching for Apsidal Precession in HD 80606 b	Priyashkumar Mistry Towards Precision Radial Velocities with Veloce		
15:30	Afternoon tea	Afternoon tea	Excursion (followed by conference dinner in the evening)	Afternoon tea		
15:45						
16:00						
16:15	Chair: Alexander Heger Holger Baumgardt The Origin of Multiple Stellar Populations in Globular Clusters	Chair: Thavisha Dharmawardena Simon Murphy Beyond isochrones: accurate stellar properties and realistic uncertainties for early type stars from stellar population synthesis.		Chair: Taïssa Danilovich David Pontin Dynamics of thermally-driven stellar winds generated in the stellar atmosphere		
16:30	Michael Albrow oMEGACat BH-2: Black-hole binary or microlensing event?	Dennis Stello Estimating fundamental stellar properties on a grand scale		Laura Driessen Stellar science with the Sydney Radio Star Catalogue		
16:45	Heather Sinclair-Wentworth Comparing the anti-correlations in NGC4372 and NGC6218	Rachael Errington Raiders of the Lost Isotopes: Hunting for the Chemical Secrets of W Aquilae		Liam Russell Understanding the Main Sequence Evolution of Intermediate Mass Stars through the Pleiades		
16:55	Mitchell Hoymans Understanding Black Hole Formation Using Dense Star Clusters	Conaire Deagan Mapping stellar surfaces via simulation-based inference using photometric and astrometric signals		Saakshi Wadhwa Confidently wrong: when short TESS baselines fail to resolve pulsation modes		
17:00	Margo Thornton A Black Hole Hiding in a Binary: A Dark Tertiary Companion to a Nearby Eclipsing Binary	Sabine Bolwell Red giant asteroseismology of Kepler background stars		Abhinav Govindan Iyer Linking Changes in Magnetic Energy to Internal and External Sources		
17:15	Nimantha Samararatunge Mudiyanseelage Mass and metallicity thresholds for the stability of monolithic supermassive stars	Muhammad Fajrin A Pulsation Analysis of the Young δ Scuti Star HD 172555 using TESS Photometry and Echelle Spectroscopy		Oliver White Spacecraft phase scintillation as a probe of solar wind activity		